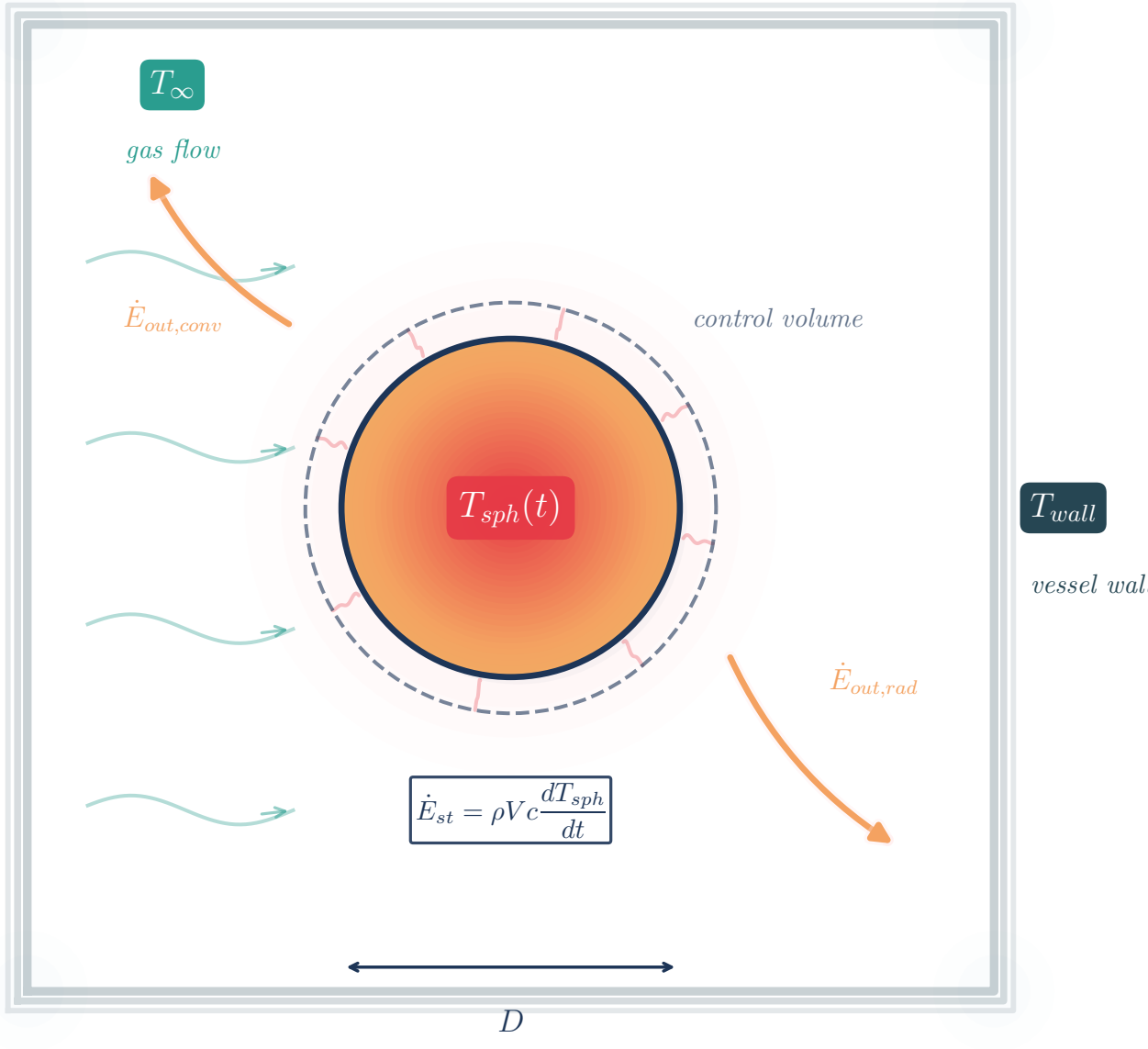
T_∞
gas flow
$\dot{E}_{out,conv}$
control volume
$T_{sph}(t)$
T_{wall}
vessel walls
$\dot{E}_{out,rad}$
$\dot{E}_{st} = \rho V c \frac{dT_{sph}}{dt}$
D